Sensing-Aided FSO Beam Tracking for UAV-Based QKD Systems

Duy-Tuan Dao*, Duy N. Luong*, Hoang D. Le†, Tuyen Ngoc Le‡§, and Anh T. Pham†
*The University of Danang-University of Science and Technology, Da Nang 50000, Vietnam
†Computer Communications Laboratory, The University of Aizu, Aizuwakamatsu 965-8580, Japan
‡Department of Electronic Engineering, Ming Chi University of Technology, New Taipei City 24301, Taiwan
§Center for Reliability Engineering, Ming Chi University of Technology, New Taipei City 24301, Taiwan

***Abstract*—This paper addresses sensing-aided tracking and non-mechanical beam steering for quantum key distributions (QKD). Notably, we propose a lightweight yet high-accuracy sensing-aided tracking for unmanned aerial vehicle (UAV)-based free-space optics (FSO)/QKD systems. By incorporating the sensing time constraint, we provide an analytical framework to evaluate the quantum bit error rate (QBER) and the secret key rate (SKR) of the decoy-state BB84 protocol over a realistic quantum channel. Also, we aim to optimize SKR performance under different UAV hovering-induced beam misalignments. To this end, we develop an optimization framework to maximize SKR by allocating optimal sensing time resources. Simulation results confirm the effectiveness of the proposed sensing-aided tracking solution and support the optimal sensing time allocation.**

***Index Terms*—Free-space optics, quantum key distribution, unmanned aerial vehicle, sensing-aided tracking systems.**

I. Introduction

Recently, unmanned aerial vehicles (UAVs) have been widely considered for critical applications, such as national defense and surveillance [1]. To enable high-speed data transmission, free-space optics (FSO) is a pragmatic solution for UAV communication systems [2]. However, the rapid advancement of quantum computing poses an urgent threat to the asymmetric cryptography systems upon which current communications rely. This necessitates secure UAV communications against quantum threats [3]. Quantum key distribution (QKD), which leverages the principles of quantum mechanics, has emerged as a promising solution to this security threat [4], [5].

Practical QKD deployment uses single photons as data carriers, for which narrow beams are required [4], [6]. The existing acquisition, tracking, and pointing (ATP) systems are employed to direct the transmitted signal and align the receiver [7]. Nevertheless, such ATP systems require heavy mechanical devices to be installed at both the transmitter and receiver [8], making them impractical for Size, Weight, and Power (SWaP)-constrained UAVs. Over the years, modulating retro-reflectors (MRRs) have been proposed as a lightweight alternative. Extensive theoretical and experimental studies [9]–[13] have demonstrated that MRRs are robust against pointing errors in FSO communication.

Nevertheless, extending MRR technology to QKD systems, as attempted in [9], [11], faces two critical challenges. First, to prevent photon-number splitting (PNS) attacks, the reflected signal must be strictly controlled by compensating for channel loss by emitting a strong interrogating power to maintain a mean photon number below one [9], [11]. Achieving this precision in highly dynamic FSO conditions with mobile UAVs requires considerable effort. Secondly, while a ground transmitter could use a broader beam to mitigate pointing errors, the inherent vibrations of the UAV induce remarkable Angle-of-Arrival (AoA) fluctuations at the ground station [10]. As demonstrated by Dabiri *et al.* [4], AoA fluctuations severely degrade QKD performance. Furthermore, practical MRR implementations still require a gimbal on the UAV for initial alignment and stabilization [10], reducing the lightweight advantage of the MRR-based system.

Given these limitations, a lightweight tracking system that enables one-way transmission is highly desirable. Fortunately, non-mechanical beam-steering optical devices, e.g., liquid crystal polarization gratings (LCPGs), spatial light modulators (SLMs), variable-focus lenses (VFL), and light detection and ranging (LIDAR) technology can provide efficient coarse and fine tracking. Based on this concept, a one-way conventional FSO system from a UAV to a ground station has been proposed in [2]. Such sensing-aided tracking systems are lighter, consume less power, and achieve performance comparable to conventional ATP-based systems. Unlike most existing works that evaluate the tracking and communication subsystems independently, the work in [2] jointly incorporates the tracking performance into the communication framework, allowing for the optimization of the sensing time. However, the beam-steering devices employed in [2] may alter the polarization states of passing photons, making this system unsuitable for polarization-encoded QKD. Consequently, a lightweight, sensing-assisted system that able to preserves quantum states has not yet been proposed, nor has its performance been analyzed under the constraint of sensing time allocation.

It is worth noting that accurately evaluating such a QKD system requires a specific channel model [4]. Pointing error models used in conventional FSO often assume that the beam waist is much larger than the receiver's aperture. However, QKD requires a narrow beam, implying that this assumption is invalid for QKD. To address this, a novel grid-based model was proposed in [4] to accurately capture the spatial intensity profile of a Gaussian beam, which is strongest at the center and decays toward the edges. Using this model, the authors analyzed the quantum bit error rate (QBER) and raw key rate[1] under a composite FSO channel. The performance

[1]Representing the rate of SPAD clicks, not the final secret key rate.

analysis in [4] remains incomplete, as it primarily considers environmental noise photons while neglecting the impact of other hardware imperfections, which may yield an optimistic evaluation.

This study, therefore, addresses the design and performance of sensing-aided tracking and non-mechanical beam steering for UAV-based FSO/QKD systems. The main contributions are summarized as follows:

- *First*, it is a proposal of a lightweight sensing-aided tracking and non-mechanical beam steering for UAV-based FSO/QKD systems
- *Secondly*, we provide an analytical framework, incorporating the tracking parameter, in terms of QBER and secret key rate (SKR) for the decoy-state BB84 protocol over quantum channels.
- *Finally*, we present an optimization strategy for SKR by determining the optimal sensing time resource allocation.

The remainder of this paper is organized as follows. Section II presents system and channel models. In Section III, we analyze QKD performance, and propose SKR optimizing method. Section IV provides simulation results, followed by conclusions in Section V.

II. System and Channel Models

A. System Descriptions

The proposed system employs a time-division multiplexing mechanism over a working period $T = t_{\rm s}+t_{\rm c}$, operating in two sequential phases: sensing $t_{\rm s}$ and QKD-based key exchange $t_{\rm c}$. Initially, a camera is utilized to estimate the receiver's position [7], [14]. The sensing process is initiated for a duration of $t_{\rm s}$. The laser beam is expanded by a beam expander (BE), and then passes through an SLM[2] and LCPGs to be split into multi-line beams and steered towards the receiver. Once reaching the receiver, a retro-reflector reflects the optical signals back to the transmitter. A photo-diode array (PD-array) at the transmitter captures these reflected beams, enabling the system to properly estimate position of the receiver (refer to LiDAR) [2].

Once the sensing phase is completed, the system transitions to the QKD phase for a duration of $t_{\rm c} = T - t_{\rm s}$. During this phase, the quantum signals are processed through the modulation and physical security modules prior to transmission.To guarantee protocol security, the system deploys two strategies. First, the two-decoy state method is implemented to mitigate photon-number splitting (PNS) attacks [11]. Second, a quantum random number generator (RNG) is employed to ensure randomness, protecting the system against RNG attacks [16]. Furthermore, to transmit these polarization-encoded photons to the receiver while preserving their polarization states, we propose a hybrid tracking system consisting of a Fast Steering Mirror (FSM) andD a Variable Focus Lens (VFL) system. Specifically, as demonstrated in two BB84 experiments utilizing polarization encoding, due to its high angular resolution, FSM is used for the fine steering stage [17],

[2]SLM enable independent and simultaneous control of multiple beams [15]

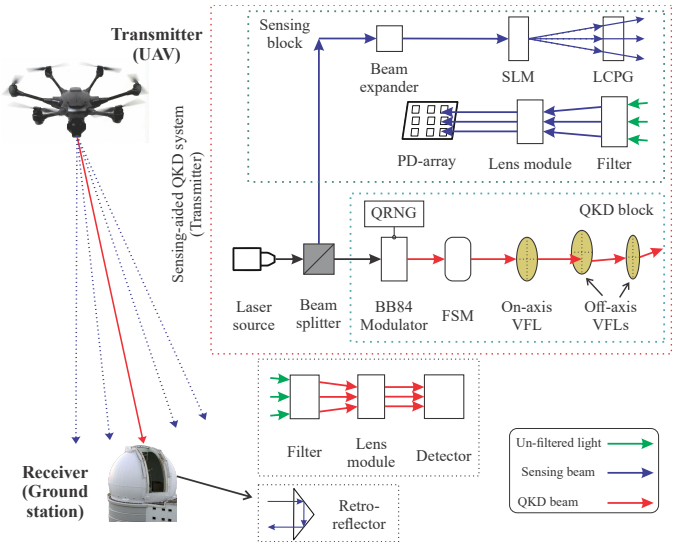


Fig. 1. The proposed sensing-aided UAV-based QKD/FSO systems.

[18]. Following the fine tracking, the beam passes through the VFL system for wide-angle coarse steering [19], [20]. This VFL architecture consists of three lenses. The first lens is aligned on-axis and is utilized to collimate the beam and adjust its divergence. The subsequent two lenses are decentered (off-axis) and are used to steer the beam in horizontal and vertical directions by adjusting their curvatures, i.e. focal lengths[3].

B. Channel Model

The total channel transmittance is given as $\tau = I_{\rm l} I_{\rm a} \tau_{\rm pg} \tau_{\rm FoV} \eta$, where $I_{\rm l}$ represents the atmospheric constant loss, $I_{\rm a}$ is the random intensity fluctuations caused by atmospheric turbulence, $\tau_{\rm pg}$ is the geometric and misalignment loss (GML), $\tau_{\rm FoV}$ characterize AoA fluctuations, and η represents detector efficiency. Their detailed description are as follows.

1) Atmospheric attenuation: There are two mechanisms impairing the signal, i.e. absorption and scattering. For the near-infrared wavelength considered, there are two main effects causing photons loss, which are Rayleigh, and Mie scattering. Specifically, scattering by molecular, small particles (Rayleigh), and aerosols, larger particles (Mie) attenuate the signal, this attenuation is modeled by the Beer-Lambert law $I_{\rm l} = \exp(-Z\beta)$ [9], where Z is the link distance, $\beta = \frac{\beta_{\rm dB}}{10^4 \log_{10} e}$ is the atmospheric extinction coefficient.

2) Atmospheric turbulence: The random variations of refractive-index along the photons propagation result in fluctuations of the received signal, i.e. fading. The parameter charactizie these fluctuations is the Rytov variance $\sigma_{\rm R}^2$, which is a function of GS height $Z_{\rm hg}$, elevation angle $\theta_{\rm elv}$, wind speed V, and refractive index structure $C_{\rm n}^2(0)$ as in [22, (12)]. In the considered scenario, since $\sigma_{\rm R}^2 < 0.3$ which implies a

[3]The liquid lenses do not alter the polarization states of the passing photons (e.g., the Optotune EL-10-30 lens [21]).

weak turbulence conditions [23], we model this factor by a log-normal distribution [9]

$$f_{I_{\rm a}}(I_{\rm a}) = \frac{1}{I_{\rm a}\sqrt{2\pi\sigma_{\rm R}^2}} \exp\left[-\frac{\left(\ln I_{\rm a} + \frac{\sigma_{\rm R}^2}{2}\right)^2}{2\sigma_{\rm R}^2}\right]. \quad (1)$$

3) Geometric and Misalignment Loss: The proposed system employs a multi-line scanning strategy to locate the receiver within a scanning range of $\varphi_{\rm r}$. In this work, we consider the pointing error to be jointly affected by the sensing error and the inherent fluctuation of the UAV.

The sensing error originates from the angular resolution of the scanning process. Specifically, the sensing error is given by $\Delta\varphi_{\rm r} = \frac{\varphi_{\rm r}}{w_{\rm s} t_{\rm s}}$ [2], where $w_{\rm s}$ denotes the scanning rate and $t_{\rm s}$ is the sensing time. The inherent vibrations of the UAV cause the fluctuations in the direction of the transmitted beam, which can be modeled as a Gaussian random variable $\theta_{\rm f} \sim \mathcal{N}(0, \sigma_{\rm f}^2)$, where the standard deviation $\sigma_{\rm f}$ represents the severity of the vibrations. Consequently, the total angular error can be modeled as $\theta_{\rm e} = \Delta\varphi_{\rm r} + \theta_{\rm f}$ [2, (23)]. GML arises from the beam overfilling effect and the radial displacement $(r_{\rm d})$ of the beam center relative to the that of the receiver's aperture, where the radial displacement is calculated as $r_{\rm d} = Z|\sin\theta_{\rm e}| \approx Z|\theta_{\rm e}|$ [2].

For a given $r_{\rm d}$, the probability of successfully capturing a photon within the receiver's aperture is evaluated using the grid-based model as [4, (17)]

$$\tau_{\rm pg}(r_{\rm d}) \approx \sqrt{\frac{2}{\pi}} \frac{\Delta x}{w_{\rm z}} \sum_{i=1}^{N_{\rm g}} \exp\left(-2\frac{(x_i - r_{\rm d})^2}{w_{\rm z}^2}\right) \operatorname{erf}\left(\frac{\sqrt{2(r_{\rm g}^2 - x_i^2)}}{w_{\rm z}}\right), \quad (2)$$

where $w_{\rm z} = w_0\sqrt{1 + \left(\frac{\lambda Z}{\pi w_0^2}\right)^2}$ is the beam radius at distance Z, w_0 is the initial beam waist, $r_{\rm g}$ is the receiving aperture's radius, which is equally divided by $N_{\rm g}$ segments with the size of $\Delta x = \frac{2r_{\rm g}}{N_{\rm g}}$, and x_i is the center of the ith segment.

4) Aperture Coupling Efficiency: Once reaching the receiver's aperture, only photons with a suitable AoA, denoted by $\theta_{\rm AoA}$, can fall into the field-of-view (FoV) of the single-photon avalanche diode (SPAD), i.e., $\theta_{\rm AoA} \leq \theta_{\rm FoV}$. The AoA is defined as deviation angle of the incoming beam and normal of the receiving aperture, which caused by error in the initial aligning process, and fluctuations at both ends during the working time. For this reason, we assume the correlation of total transmitting angular deviation $\theta_{\rm e}$ and angle of arrival $\theta_{\rm AoA}$ is weak. Therefore AoA fluctuation can be modeled as an independent Gaussian random variable, i.e. $\theta_{\rm AoA} \sim \mathcal{N}(0, \sigma_{\rm AoA}^2)$ [4], [24]. Let $\tau_{\rm FoV}$ denote the probability of successfully coupling a photon into the detector, its probability density function (PDF) can be given as [4, (35)]

$$f_{\tau_{\rm FoV}}(\tau_{\rm FoV}) = P_0\delta(\tau_{\rm FoV}) + P_1\delta(\tau_{\rm FoV} - 1), \quad (3)$$

where $P_0 = \exp\left(-\frac{\theta_{\rm FoV}^2}{2\sigma_{\rm AoA}^2}\right)$, $P_1 = 1 - P_0$, and $\delta(\cdot)$ denotes the Dirac delta function.

III. Performance Analysis & Optimization

A. Performance Analysis

1) Quantum Bit Error Rate (QBER): Average QBER over channel transmittance τ is expressed by [25, (14)]

$$\langle E_\mu \rangle = \frac{\langle E_\mu Q_\mu \rangle}{\langle Q_\mu \rangle} = \frac{\int_0^\infty E_\mu Q_\mu(\tau) f_\tau(\tau) {\rm d}\tau}{\int_0^\infty Q_\mu(\tau) f_\tau(\tau) {\rm d}\tau}, \quad (4)$$

where $\langle E_\mu Q_\mu \rangle$, and $\langle Q_\mu \rangle$ denote the average probability of erroneous detections, and raw detection, respectively. Conditional on a channel transmittance τ, $E_\mu Q_\mu(\tau)$ is the probability of erroneous detections per pulse, given as [25, (15)]

$$E_\mu Q_\mu(\tau) = e_0 p'_{\rm dark} + (e_{\rm pol} + e_0 P_{\rm AP})(1 - e^{\mu\tau}), \quad (5)$$

where $p'_{\rm dark} = p_{\rm dark}(1 + P_{\rm AP})$, $p_{\rm dark}$ is the dark count rate $P_{\rm AP}$ is the after pulsing probability caused by the ultra-fast rate, e_0 is the error probability caused by noise photon detection events, and $e_{\rm pol}$ is the error rate due to polarization error. Similarly, $Q_\mu(\tau)$ quantifies the probability of a detection events, and is expressed as [25, (16)]

Proposition 1. *Average QBER over the fluctuating channel can be quantified as*

$$\langle E_{\mu_n} \rangle = \frac{\langle E_{\mu_n} Q_{\mu_n} \rangle}{\langle Q_{\mu_n} \rangle} = \frac{e_0 p'_{\rm dark} + \left(e_{\rm pol} + e_0 P_{\rm AP}\right)\left[1 - G(\mu_n)\right]}{p'_{\rm dark} + (1 + P_{\rm AP})\left[1 - G(\mu_n)\right]}, \quad (6)$$

in which, the function $G(z)$ is expressed as

$$G(z) = P_0 + \frac{P_1}{\sqrt{\pi}} \sum_{i=1}^{M} w_i \exp\left(\frac{\lambda_{\rm IG,i}}{\mu_{\rm IG,i}}\left(1 - \sqrt{1 + \frac{2z\mu_{\rm IG,i}^2}{\lambda_{\rm IG,i}}}\right)\right), \quad (7)$$

where w_i and x_i denote the Gauss-Hermite weights and nodes, respectively. Also, $\mu_{\rm IG,i}$, and $\lambda_{\rm IG,i}$ are updated based on $\theta_{{\rm e},i}$, where $\theta_{{\rm e},i} = \sqrt{2}\sigma_{\rm f} x_i + \Delta\varphi_{\rm r}$.

Proof. See Appendix A. □

Remark 1. *The approximation used in* (7) *quickly converges to its exact form expression with $M = 50$ terms*

2) Average Secret Key Rate (SKR): Since the system operates in two phases, in which the first phase is for sensing (t_s), and only the second phase is used for QKD conducting. Using the decoy-state BB84 protocol, the information-theoretic lower limit for SKR is given as [9, (12)]

$$\mathcal{S}\mu \geq \frac{T - t_s}{T}\mathcal{R}spd\Big\{-\langle Q_\mu\rangle f_{\rm e} H_2(\langle E_\mu\rangle) + \langle Q_1^L\rangle\big[1 - H_2\big(\langle e_1^U\rangle\big)\big]\Big\}, \quad (8)$$

where $\mathcal{R}$, s, p, d, $f_{\rm e}$, and $H_2(\cdot)$ denote repetition rate, sifting coefficient, parameter estimation coefficient, fraction of signal pulses, reconciliation efficiency, and Shannon's entropy function, respectively. Also, $\langle Q_1^L\rangle$, $\langle e_1^U\rangle$ could be found at [9, (13)], and [9, (14)], while Y_1^L is found in [26, (G9)].

B. SKR Performance Optimization

The impact of UAV hovering-induced misalignment can be alleviated by sensing-aided tracking. In other words, longer sensing time of $t_{\rm s}$ enhances the pointing accuracy. This, in turn, reduces SKR performance due to the shorter time dedicated to the QKD phase. This trade-off indicates the existence of an optimal $t_{\rm s}$. To this end, this study aims to maximize the SKR performance by allocating the optimal sensing time resource of $t_{\rm s}$.

Remark 2. *The SKR defined in* (8) *incorporates several complicated functions, i.e., the Shannon entropy function, the error function, and Gauss-Hermite quadratures. Therefore, directly maximizing this SKR using* (8) *is analytically intractable and requires extensive computational resources.*

We propose a computationally efficient surrogate objective function capable of capturing the competing factors of this trade-off, i.e.,

$$\mathcal{P}: \max_{t_{\rm s}} \quad F(t_{\rm s}) = \frac{T - t_{\rm s}}{T} \mathbb{E}\left[\tau_{\rm pg}(\tau_{\rm s})\right], \tag{9a}$$

$$\text{s.t.} \quad 0 \le t_{\rm s} < T, \tag{9b}$$

where $\mathbb{E}\left[\tau_{\rm pg}(t_s)\right]$ is given as

$$\mathbb{E}\left[\tau_{\rm pg}(t_s)\right] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^{M} w_i \tau_{\rm pg}(\theta_{{\rm e},i}). \tag{10}$$

The proposed surrogate function isolates two critical components, i.e., the time-penalty factor of $\frac{T-t_{\rm s}}{T}$ and the expected GML of $\mathbb{E}[\tau_{\rm pg}(t_{\rm s})]$. As $t_{\rm s}$ increases, the time-penalty factor strictly decreases, whereas $\mathbb{E}[\tau_{\rm pg}(t_{\rm s})]$ strictly increases[4]. Consequently, this surrogate function can be assumed to be unimodal in the considered scenario. To this end, Brent's method, which is summarized in Algorithm 1, emerges as an excellent candidate to reliably locate the optimal $t_{\rm s}$ [27].

IV. Numerical Results & Discussions

This section evaluates the performance of the proposed system over the FSO channels[5] Parameters are listed in Table I, and Monte Carlo simulations with 10^6 samples are used to validate the analytical results.

[4]Note that when $t_{\rm s} \to \infty$, $\mathbb{E}[\tau_{\rm pg}(t_{\rm s})]$ will reach its ceiling, whose value is defined by the UAV fluctuations $\sigma_{\rm f}$.

[5]A key distinction of this work is the integration of the sensing time into the UAV-to-ground QKD performance analysis. Recent works, such as [4], consider similar scenarios but neglect this critical constraint. Consequently, comparing our time-constrained framework with those models would be inequitable. Furthermore, though MRR-aided QKD systems are proposed in [11], [13], their evaluation relies on simplified channel conditions, which require extensive effort to be comprehensively analyzed. A more comprehensive comparison would be in our future work.

Algorithm 1: Brent Optimization for Solving $\mathcal{P}$

Input: Time bounds $[a, b] = [\epsilon, T - \epsilon]$, GH nodes/weights $\{\mathbf{x}[i], \mathbf{w}[i]\}_{i=1}^{M}$
Output: Optimal sensing time t_s^*

Initialize state variables $x, w, v = a + \frac{3-\sqrt{5}}{2}(b-a)$;
Initialize previous step size $e = 0$;
Calculate initial objective $F(x)$ as in (9a);
while $\frac{1}{2}(b-a) > \epsilon$ **do**
  Calculate trial point $t_{\rm SPI}$ via Successive Parabolic Interpolation (SPI) using (x, w, v);
  if $t_{\rm SPI} \in (a, b)$ ***and*** $|t_{\rm SPI} - x| < \frac{1}{2}|e|$ **then**
    $t_s = t_{\rm SPI}$;
  else
    $t_s = \text{GoldenSection}(a, b, x)$;
  end if
  $e = t_s - x$;
  Calculate surrogate value $F(t_s)$ as in (9a);
  Shorten search interval $[a, b]$ based on Golden Section Search's rule to bound the maximum based on $F(t_s)$ and $F(x)$;
  Re-order x, w, v as the 3 best evaluations;
end while
return x;

TABLE I
System Parameters [2], [4], [12], [25]

Symbol	Value	Symbol	Value
Ground Station (Bob)			
$Z_{\rm hg}$	10 m	e_0	0.5
$\theta_{\rm FoV}$	100 μrad	$e_{\rm pol}$	0.033
$r_{\rm g}$	8 cm	$p_{\rm dark}$	10^{-4}
$P_{\rm AP}$	2 %		
UAV (Alice)			
$\theta_{\rm elv}$	60°	$\varphi_{\rm r}$	1 rad
λ	1550 nm	$w_{\rm s}$	1.8×10^4 points/s
w_0	10 cm	T	1 s
$\sigma_{\rm f}$	200 μrad		
Quantum Channel			
Z	1 km	$\sigma_{\rm AoA}$	50 μrad
V	5 m/s	$\beta_{\rm dB}$	0.43 dB/km
$C_{\rm n}^2(0)$	10^{-14} m$^{-2/3}$	η	0.6
BB84 Protocol and others			
$\mu_{\rm n}$	0.3	$f_{\rm e}$	1
$\nu_{\rm n}$	0.09	s	0.5
$\mathcal{R}$	10^9 pulses/s	p	0.75
d	0.5	$N_{\rm g}$	10

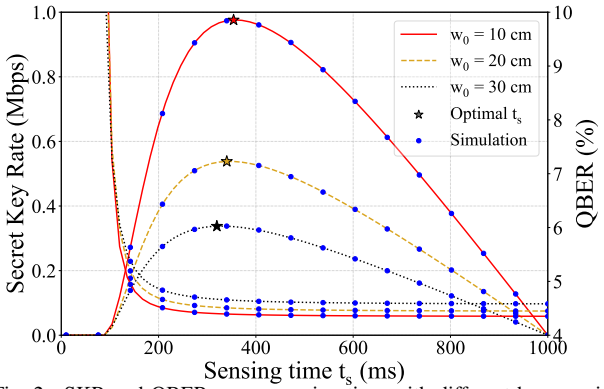


Fig. 2. SKR and QBER across sensing time with different beam waist.

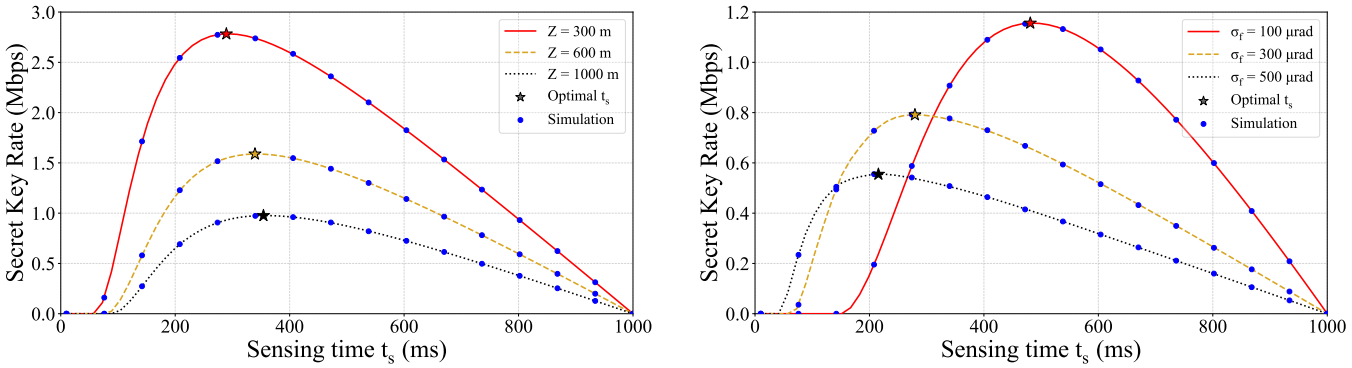


Fig. 3. Impact of sensing time allocation on the SKR and optimal t_s selections under different (a) Link length, (b) UAV fluctuations.

TABLE II
RUNTIME AND ACCURACY COMPARISON (EVALUATED AT $w_0 = 10$ CM, $Z = 1000$ M)

Optimization Algorithm	Objective Function	Runtime (ms)	Optimal t_s (ms)	Achieved SKR (Mbps)
Exhaustive Search (Baseline)	Gauss-Hermite based SKR	37841	357.22	0.97615 (100%)
Brent's Method	Gauss-Hermite based SKR	180	357.26	0.97615 (100%)
Exhaustive Search	Surrogate	20454	354.16	0.97608 (99.99%)
Brent's Method (Proposed)	**Surrogate**	**26.5**	**354.2**	**0.97608 (99.99%)**

The optimal t_s values determined by Brent's optimization are denoted by star markers, with colors corresponding to their respective curves. As seen, the SKR curves are unimodal, and highly dependent on the sensing time t_s. Notably, the theoretical optimal t_s markers closely coincide with the peaks of the SKR curves. Furthermore, Monte Carlo simulations show a strong agreement with the analytical lines in all scenarios, validating the correctness of the proposed mathematical framework.

Figure 2 highlights the importance of selecting a proper beam waist and sensing time. When the sensing time is below 200 ms, large pointing errors occur nearly disable QKD transmission. As t_s exceeds 200 ms, the QBER curves become more stable indicating that enhancing the channel cannot tolerate the impact of hardware limitations. At the same time, the SKR increases to a peak when the channel transmittance improves. After that, the SKR decreases because a long sensing duration reduces the available time for the quantum phase. Moreover, the results confirm the necessity of using a narrow beam in QKD as a broader beam reduces misalignment errors but increases geometric losses. Specifically, the maximum SKR for a narrow beam with $w_0 = 10$ cm reaches close to 1 Mbps, while a broader beam at $w_0 = 30$ cm achieves a peak of under 0.4 Mbps. An increase in w_0 enlarges the beam footprint and makes the system more robust against angular deviations. The optimal t_s shifts to the left. This shows that the system will conduct less sensing and allocate more time to the QKD phase to maximize the final SKR.

Table. II compares Brent's method against the exhaustive search (baseline) in terms of runtime and accuracy. For the exhaustive search, 10 000 t_s samples are generated within the range of 10 ms to 1000 ms. When operating on the Gauss-Hermite based function, Brent's method reduces the runtime compared to the baseline exhaustive search while achieving approximately identical accuracy. Furthermore, applying Brent's method to the proposed surrogate function further accelerates the computation by approximately 7 times compared to its execution on the Gauss-Hermite function. Notably, the optimal t_s derived from the surrogate model exhibits only a slight deviation ($\approx$ 3 ms) compared to the Gauss-Hermite model, yielding an approximate SKR retention of 99.99%.

Figure 3a illustrates the impact of increasing link distance Z on the system performance. An increase in Z reduces the received signal power due to atmospheric attenuation, and geometric misalignment loss. Specifically, a long transmission distance causes any given transmitting angle error to create a larger radial displacement (r_d) at the receiver. To mititage this impairment, the system must increase sensing accuracy by allocating a larger time amount for the sensing process. For this reason, the optimal t_s increases at longer distances.

Figure 3b illustrates the severe impact of UAV hovering fluctuations (σ_f) on the SKR and the optimal sensing time allocation. Predictably, when the UAV vibrations become more intense (i.e., σ_f increases from 100 μrad to 500 μrad), the maximum achievable SKR drops significantly. Also, the figure reveals a counter-intuitive behavior regarding resource allocation, under highly unstable conditions (larger σ_f), the optimal sensing time t_s notably shifts to the left. Specifically, the optimal t_s decreases from approximately 480 ms at $\sigma_f = 100$ μrad to around 210 ms at $\sigma_f = 500$ μrad. This adaptation occurs because, under severe inherent vibrations, the pointing accuracy is strictly bounded by the hardware instability. In such scenarios, prolonging the sensing duration marginally enhance tracking performance. Consequently, the

system adapts by allocating more time for the QKD phase.

V. Conclusions

This paper has comprehensively investigated the performance of a time-constrained sensing-assisted FSO/QKD system over realistic FSO quantum channels. We newly derived mathematical expressions for key performance metrics, i.e. the QBER and SKR of the decoy-state BB84 protocol. Furthermore, an optimization strategy to maximize the system's performance via optimal sensing time allocation was provided. The accuracy of the proposed analytical framework was rigorously verified through Monte Carlo simulations. The numerical results offer meaningful insights into system configuration, highlighting the crucial trade-off between tracking accuracy and quantum transmission time. Moreover, our contributions can be expanded for one-way dynamic QKD deployment. Future work will extend this framework to investigate dynamic adaptive beam waist optimization and include comprehensive comparisons against alternative schemes, e.g. MRR-aided systems.

Appendix A
Proof of Proposition 1

Evaluating (4) requires the following function $G(z) = \int_0^\infty e^{-\mu\tau} f_\tau(\tau)\mathrm{d}\tau$. To solve this function, we first conditioning on a fixed θ_e, which results in a deterministic radial displacement r_d. Under this condition, the geometric misalignment loss in (2) becomes a constant $\tau_\mathrm{pg}(\theta_\mathrm{e})$. The composite channel τ collapses to $\tau = \tau'\tau_\mathrm{FoV}$, where $\tau' = \tau_\mathrm{C}\tau_\mathrm{l}$ and $\tau_\mathrm{C} = \tau_\mathrm{l}\tau_\mathrm{pg}(\theta_\mathrm{e})$ follows a log-normal distribution. According to [28, (Sec. II)], the probability distribution of $\tau = \tau'\tau_\mathrm{FoV}$ is approximated by an Inverse Gaussian distribution $f_{\tau|\theta_\mathrm{e}}(\tau \mid \theta_\mathrm{e}) = P_0\delta(\tau) + P_1\mathrm{IG}(\tau(\theta_\mathrm{e}); \mu_\mathrm{IG}, \lambda_\mathrm{IG})$, where $\mu_\mathrm{IG} = \tau_\mathrm{C}$ and $\lambda_\mathrm{IG} = \frac{\tau_\mathrm{C}}{\exp(\sigma_\mathrm{R}^2)-1}$.

The total angular error follows a Gaussian distribution, i.e. $\theta_\mathrm{e} \sim \mathcal{N}(\Delta\varphi_\mathrm{r}, \sigma_\mathrm{f}^2)$. Applying a change of variable $x = \frac{\theta_\mathrm{e}-\Delta\varphi_\mathrm{r}}{\sqrt{2}\sigma_\mathrm{f}}$, $G(z)$ can be expressed as

$$G(z) = P_0 + \int_{-\infty}^{\infty} P_1 \exp\left(\frac{\lambda_\mathrm{IG}(x)}{\mu_\mathrm{IG}(x)}\left(1 - \sqrt{1 + \frac{2z\mu_\mathrm{IG}^2(x)}{\lambda_\mathrm{IG}(x)}}\right)\right) \times \frac{1}{\sqrt{2\pi}\sigma_\mathrm{f}} \exp\left(-x^2\right)\sqrt{2}\sigma_\mathrm{f}\mathrm{d}x. \quad (11)$$

applying Gauss-Hermite quadrature, (11) is given as in (7).

References

[1] W. Zhao *et al.*, "A survey on DRL-based UAV communications and networking: DRL fundamentals, applications and implementations," *IEEE Communications Surveys & Tutorials*, 2025.

[2] Y. Li *et al.*, "Sensing assisted optical wireless communication for uavs," *IEEE Transactions on Vehicular Technology*, vol. 73, pp. 18 620–18 634, 12 2024.

[3] H. Luo *et al.*, "ESCM: An efficient and secure communication mechanism for uav networks," *IEEE Transactions on Network and Service Management*, vol. 21, no. 3, pp. 3124–3139, 2024.

[4] M. T. Dabiri *et al.*, "A unified framework for uav-based free-space quantum links: Beam shaping and adaptive field-of-view control," 2025.

[5] R. Renner and R. Wolf, *Perspective on the QKD Versus PQC Debate*. Cham: Springer Nature Switzerland, 2026, pp. 163–170.

[6] P. V. Trinh, S. Sugiura, C. Xu, and L. Hanzo, "Toward quantum sagins harnessing optical riss: Applications, advances, and the road ahead," *IEEE Network*, vol. 39, no. 3, pp. 215–222, 2025.

[7] Y. Kaymak *et al.*, "A survey on acquisition, tracking, and pointing mechanisms for mobile free-space optical communications," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 2, pp. 1104–1123, 2018.

[8] R. Abdelfatah *et al.*, "A review on pointing, acquisition, and tracking approaches in uav-based fso communication systems," *Optical and Quantum Electronics*, vol. 54, no. 9, p. 571, 2022.

[9] D. N. Luong, D.-T. Dao, C. T. Nguyen, H. D. Le, and A. T. Pham, "Modulating retroreflector-aided uav-based FSO/QKD systems," *IEEE Communications Letters*, pp. 1–1, 2026.

[10] P. V. Trinh, A. Carrasco-Casado, T. Okura, H. Tsuji, D. R. Kolev, K. Shiratama, Y. Munemasa, and M. Toyoshima, "Experimental channel statistics of drone-to-ground retro-reflected fso links with fine-tracking systems," *IEEE Access*, vol. 9, pp. 137 148–137 164, 2021.

[11] W. S. Rabinovich, R. Mahon, M. S. Ferraro, P. G. Goetz, M. Bashkansky, R. E. Freeman, J. Reintjes, and J. L. Murphy, "Free space quantum key distribution using modulating retro-reflectors," *Opt. Express*, vol. 26, no. 9, pp. 11 331–11 351, Apr 2018. [Online]. Available: https://opg.optica.org/oe/abstract.cfm?URI=oe-26-9-11331

[12] M. T. Dabiri *et al.*, "Modulating retroreflector based free space optical link for UAV-to-ground communications," *IEEE Transactions on Wireless Communications*, vol. 21, no. 10, pp. 8631–8645, Oct. 2022.

[13] M. Zhu *et al.*, "Free-space qkd with modulating retroreflectors based on the b92 protocol," *Entropy*, vol. 24, no. 2, 2022.

[14] Y. Li, L. Wu, J. Cheng, J. Dang, B. Zhu, X. Zhang, and Z. Zhang, "Ear analysis of integrated optical wireless communication and sensing scheme for hovering uavs," *IEEE Transactions on Vehicular Technology*, pp. 1–12, 01 2026.

[15] Y. Jian *et al.*, "Optical beam steerable orthogonal frequency division multiplexing (ofdm) non-orthogonal multiple access (noma) visible light communication using spatial-light modulator based reconfigurable intelligent surface," *IEEE Photonics Journal*, vol. 15, pp. 1–8, 2023.

[16] M. Herrero-Collantes and J. C. Garcia-Escartin, "Quantum random number generators," *Reviews of Modern Physics*, vol. 89, 02 2017.

[17] X.-T. Zheng, Q.-F. Zhang, J. Ling, G.-C. Guo, and Z.-F. Han, "Free-space continuous-variable quantum key distribution under high background noise," *npj Quantum Information*, vol. 11, no. 1, p. 52, 2025.

[18] W.-Y. Liu *et al.*, "Experimental free-space quantum key distribution with efficient error correction," *Opt. Express*, vol. 25, no. 10, pp. 10 716–10 723, May 2017.

[19] V. V. Mai and H. Kim, "Non-mechanical beam steering and adaptive beam control using variable focus lenses for free-space optical communications," *Journal of Lightwave Technology*, vol. 39, no. 24, pp. 7600–7608, Dec 2021.

[20] M. Zohrabi *et al.*, "Wide-angle nonmechanical beam steering using liquid lenses," *Opt. Express*, vol. 24, no. 21, pp. 23 798–23 809, Oct 2016. [Online]. Available: https://opg.optica.org/oe/abstract.cfm?URI=oe-24-21-23798

[21] Optotune Switzerland AG, *Datasheet: EL-10-30-Series Fast Electrically Tunable Lens*, Dec. 2019, revision: 10.12.2019. [Online]. Available: https://archive.optotune.com/images/products/Optotune

[22] Y. Ata and M.-S. Alouini, "Performance of integrated ground-air-space FSO links over various turbulent environments," *IEEE Photonics Journal*, vol. 14, no. 6, pp. 1–16, Dec. 2022 .

[23] Y. Ata *et al.*, "Optical wireless communication in atmosphere and underwater: Statistical models, improvement techniques, and recent applications," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 4248–4284, 01 2025.

[24] D. Singh and S. Ramabadran, "Comprehensive performance analysis of hovering uav-based fso communication system," *IEEE Photonics Journal*, vol. 14, pp. 1–13, 10 2022.

[25] Y. Takihara, H. D. Le, C. T. Nguyen, and A. T. Pham, "Qiskit-enabled simulation of quantum key distribution over starlink satellite networks," in *IEEE Int. Conf. Advanced Technol. Commun.*, 2025, pp. 1–6.

[26] D. Vasylyev, W. Vogel, and F. Moll, "Satellite-mediated quantum atmospheric links," *Phys. Rev. A*, vol. 99, no. 5, p. 053830, May 2019.

[27] D. G., *Mathematics of Computation*, vol. 28, no. 127, pp. 865–866, 1974. [Online]. Available: http://www.jstor.org/stable/2005713

[28] T. Li *et al.*, "On the optimality of inverse gaussian approximation for lognormal channel models," *IEEE Signal Processing Letters*, vol. 31, pp. 2625–2629, 2024.